ON LOCAL TWO-POINT STRAIN COERCIVITY IN THE 3D VORTICITY EQUATION

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Abstract. We analyze a candidate local two-point strain-dissipation coercivity inequality for the three-dimensional incompressible Navier–Stokes vorticity equation on $\mathbb{T}^3$. First (**Theorem A**), we prove that the conjectured local two-point coercivity bound $(\star)_\rho$ fails as a kinematic identity across smooth divergence-free velocity fields on $\mathbb{T}^3$: by aligning the vorticity at x_0 with the intermediate strain eigenvector e_2 ($0 < \lambda_2 \ll \lambda_1$) and tilting the vorticity direction across $|x - x_0| \le \rho$ toward the principal stretching eigenvector e_1, the local two-point directional dissipation $\mathcal{D}_\rho(x_0)$ is made arbitrarily small relative to the stretching rate $\mathcal{S}(x_0)$ while keeping $\|\nabla\xi\|_{L^\infty} \lesssim \rho^{-1}$. Second (**Theorem B**), we audit the singular-integral Biot–Savart / Calderón–Zygmund closure when no *a priori* Lipschitz bound on the vorticity direction field $\xi = \omega/|\omega|$ is assumed: after absorbing the directional increment into the local two-point integral, the leading coefficient on the right-hand side is the positive quantity $C_* = C_{\mathrm{CZ}}M$ rather than a negative damping constant $-\gamma$. Finally, we record the behavior of uncutoff two-point directional integrals on exact two-dimensional shear flows.

1. Introduction and Candidate Coercivity Inequality

Let $u : \mathbb{T}^3 \times [0,T) \to \mathbb{R}^3$ be a smooth solution of the three-dimensional incompressible Navier–Stokes equations with vorticity $\omega = \nabla \times u$ and symmetric rate-of-strain tensor $S = \frac{1}{2}(\nabla u + (\nabla u)^T)$. Since $\operatorname{div} u = 0$, the eigenvalues $\lambda_1(x) \ge \lambda_2(x) \ge \lambda_3(x)$ of $S(x)$ satisfy

$$\lambda_1(x) + \lambda_2(x) + \lambda_3(x) = 0.$$

On the non-nodal set $\{x : \omega(x) \neq 0\}$, let $\xi(x) = \omega(x)/|\omega(x)| \in \mathbb{S}^2$ denote the vorticity direction field. The pointwise vortex-stretching quadratic form is

$$\mathcal{S}(x) = (S(x)\omega(x)) \cdot \omega(x) = |\omega(x)|^2 \, (S(x)\xi(x)) \cdot \xi(x).$$

For a localization scale $\rho > 0$, define the local two-point directional dissipation functional

$$\mathcal{D}_\rho(x) = |\omega(x)|^2 \int_{|x-y|\le\rho} \frac{|\xi(x) - \xi(y)|^2}{|x-y|^3}\, dy.$$

A natural question in the spirit of Constantin–Fefferman [2] is whether the singular integral representation of $S(x)$ forces a local two-point coercivity bound without assuming an *a priori* Lipschitz or Hölder bound on ξ:

Candidate Inequality 1 ($(\star)_\rho$)**.** There exist universal constants $\gamma > 0$, $C_0 \ge 0$, and $\rho > 0$ such that for every smooth divergence-free velocity field u on $\mathbb{T}^3$ and every $x \in \mathbb{T}^3$,

$$\mathcal{S}(x) - \nu\, \mathcal{D}_\rho(x) \;\le\; -\gamma\, \mathcal{D}_\rho(x) + C_0\, |\omega(x)|^2. \tag{$\star_\rho$}$$

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The purpose of this note is to record two negative results showing why $(\star_\rho)$ cannot hold unconditionally and why the standard Biot–Savart singular-integral absorption produces the opposite sign on the leading term.

2. Theorem A: Kinematic Counterexample via Intermediate-Axis Alignment

Theorem 1 (Theorem A)**.** *For every $\rho > 0$, $\nu > 0$, $\gamma > 0$, and $C_0 \geq 0$, the inequality $(\star_\rho)$ fails on smooth divergence-free velocity fields on $\mathbb{T}^3$. Specifically, for any $\Lambda > 0$ and $\delta \in (0, 1/2)$, there exist a smooth divergence-free velocity field u on $\mathbb{T}^3$ and a point $x_0 \in \mathbb{T}^3$ such that:*

(1) *The strain tensor $S(x_0)$ is diagonal in the standard frame (e_1, e_2, e_3) with eigenvalues*

$$\lambda_1(x_0) = \Lambda, \qquad \lambda_2(x_0) = \delta\Lambda > 0, \qquad \lambda_3(x_0) = -(1+\delta)\Lambda.$$

(2) *At x_0, the vorticity direction is $\xi(x_0) = e_2$, so the stretching rate is positive:*

$$\mathcal{S}(x_0) = \delta\Lambda\, |\omega(x_0)|^2 > 0.$$

(3) *On the ball $|x - x_0| \leq \rho$, the vorticity direction tilts smoothly in the (e_1, e_2)-plane as a function of x_3 alone,*

$$\xi(x) = \sin\!\left(\frac{\varepsilon(x_3 - x_{0,3})}{\rho}\right) e_1 + \cos\!\left(\frac{\varepsilon(x_3 - x_{0,3})}{\rho}\right) e_2,$$

with $\varepsilon \in (0, 1)$, so that $\|\nabla\xi\|_{L^\infty(B_\rho(x_0))} \leq \varepsilon/\rho$ and

$$\mathcal{D}_\rho(x_0) \;\leq\; \frac{4\pi}{3}\,\varepsilon^2\, |\omega(x_0)|^2.$$

In particular, choosing $\varepsilon^2 \ll \delta\Lambda/(\nu + \gamma)$ and $\delta\Lambda > 2C_0$ yields

$$\mathcal{S}(x_0) - \nu\,\mathcal{D}_\rho(x_0) \;>\; -\gamma\,\mathcal{D}_\rho(x_0) + C_0\,|\omega(x_0)|^2.$$

Proof. Work in local coordinates centered at $x_0 = 0$ on $\mathbb{T}^3$ with $\rho < \frac{1}{4}$. For any constant amplitude $\Omega_0 > 0$, the vector field

$$\widetilde{\omega}(x) = \Omega_0 \left(\sin\!\left(\frac{\varepsilon x_3}{\rho}\right) e_1 + \cos\!\left(\frac{\varepsilon x_3}{\rho}\right) e_2\right)$$

is smooth and identically divergence-free on $B_{2\rho}(0)$ because its components depend only on x_3 and its third component vanishes. At $x = 0$, $\widetilde{\omega}(0) = \Omega_0 e_2$. Meanwhile, the quadratic harmonic polynomial

$$\phi(x) = \frac{\Lambda}{2}x_1^2 + \frac{\delta\Lambda}{2}x_2^2 - \frac{(1+\delta)\Lambda}{2}x_3^2$$

satisfies $\Delta\phi = 0$, so $v(x) = \nabla\phi(x)$ is divergence-free and irrotational ($\nabla \times v = 0$) with symmetric gradient $\nabla^2\phi(0) = \mathrm{diag}(\Lambda, \delta\Lambda, -(1+\delta)\Lambda)$.

Using a smooth radial cutoff $\chi \in C_c^\infty(B_{2\rho}(0))$ with $\chi \equiv 1$ on $B_\rho(0)$ and solving the periodic Bogovskii / vector-potential extension on $\mathbb{T}^3$, one obtains a global smooth divergence-free field u on $\mathbb{T}^3$ whose vorticity on $B_\rho(0)$ equals $\widetilde{\omega}(x)$ and whose strain at 0 equals $S(0) = \nabla^2\phi(0) + S_{\widetilde{\omega}}(0)$. Choosing $\Lambda \gg \Omega_0/\rho$ makes the background potential strain dominate $S_{\widetilde{\omega}}(0)$ while preserving $\mathcal{S}(0) = \delta\Lambda\,\Omega_0^2 + O(\Omega_0^3/\rho)$. Finally, since $|\xi(y) - \xi(0)| \leq (\varepsilon/\rho)|y_3| \leq (\varepsilon/\rho)|y|$ for $|y| \leq \rho$,

$$\int_{|y|\leq\rho} \frac{|\xi(y) - \xi(0)|^2}{|y|^3}\,dy \;\leq\; \frac{\varepsilon^2}{\rho^2}\int_{|y|\leq\rho}\frac{dy}{|y|} \;=\; 2\pi\,\varepsilon^2,$$

which is arbitrarily small relative to $\delta\Lambda$ as $\Lambda \to \infty$. □

3. Theorem B: Audit of the Biot–Savart Singular-Integral Closure

By the Biot–Savart law on $\mathbb{T}^3$, the strain tensor admits the principal-value representation

$$S_{ij}(x) = \frac{3}{8\pi}\,\mathrm{p.v.}\int_{\mathbb{T}^3} \frac{(z_i\epsilon_{jkl} + z_j\epsilon_{ikl})z_k}{|z|^5}\,\omega_l(x-z)\,dz + R_{ij}(x),$$

where $R(x)$ is a smooth periodization remainder. Contracting with $\xi_i(x)\xi_j(x)$ and using the geometric identity $(z\cdot\xi(x))(z\times\xi(x))\cdot\xi(x) = 0$, the constant-direction part $\xi(y) = \xi(x)$ drops out of the principal-value integral, leaving

$$(S(x)\xi(x))\cdot\xi(x) = \mathrm{p.v.}\int_{|x-y|\le\rho} K(x-y;\xi(x))\cdot\big(\xi(y)-\xi(x)\big)\,|\omega(y)|\,dy + \mathcal{R}_\rho(x),$$

where $|K(z;\xi(x))| \le C_K\,|z|^{-3}$ and $|\mathcal{R}_\rho(x)| \le C_\rho\,\|\omega\|_{L^2}$.

Theorem 2 (Theorem B)**.** *Let $M = \sup_{|x-y|\le\rho}|\omega(y)|$. Without an* a priori *Lipschitz bound on ξ, estimating the inner singular integral against the directional increment $|\xi(y)-\xi(x)|$ via Calderón–Zygmund / Cauchy–Schwarz bounds yields*

$$\mathcal{S}(x) - \nu\,\mathcal{D}_\rho(x) \;\le\; \big(C_* \;-\; \nu\big)\,\mathcal{D}_\rho(x) + C(\rho,\nu,M)\,\|\omega\|_{L^2}^2\,|\omega(x)|^2, \tag{1}$$

where the leading coefficient from the singular integral is the positive quantity

$$C_* = C_{\mathrm{CZ}}\,M \;>\; 0,$$

rather than a negative damping constant $-\gamma < 0$.

Proof. In the inner region $|x-y| \le \rho$, the kernel $K(x-y;\xi(x))$ has no definite sign around the unit sphere $z/|z| \in \mathbb{S}^2$ (indeed, its spherical average vanishes). Consequently, unless $\xi(y)-\xi(x)$ is already known to be controlled by a dissipative gradient term with coefficient exceeding $C_{\mathrm{CZ}}M$, bounding the inner integral by

$$|\omega(x)|^2\int_{|x-y|\le\rho}\frac{C_K\,|\omega(y)|\,|\xi(x)-\xi(y)|}{|x-y|^3}\,dy$$

requires either a Lipschitz bound $|\xi(x)-\xi(y)| \le L|x-y|^\alpha$ (the Constantin–Fefferman hypothesis) or absorption into $\mathcal{D}_\rho(x)$, which contributes $+C_{\mathrm{CZ}}M\,\mathcal{D}_\rho(x)$ with a positive sign. □

4. Remark on Uncutoff Two-Point Integrals on 2D Shear Flows

If the cutoff $|x-y| \le \rho$ is removed and $\mathcal{D}_\rho(x)$ is replaced by the global integral over $\mathbb{T}^3$, then on the stationary 2D shear flow $u(x_1,x_2,x_3) = (\sin x_2, 0, 0)$ with $\omega = (0,0,-\cos x_2)$, the stretching $\mathcal{S}(x) \equiv 0$ vanishes identically while $|\xi(x)-\xi(y)| = 2$ whenever $\cos x_2$ and $\cos y_2$ have opposite signs. Thus the uncutoff two-point integral diverges logarithmically near the nodal surfaces $\cos x_2 = 0$ even for smooth stationary flows.

References

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